01 General

SN DL pro 780 :	11097		
Customer:	Northwestern University		
Order Number:	VB.09.0516/01	**Production Date:**	27.03.2009

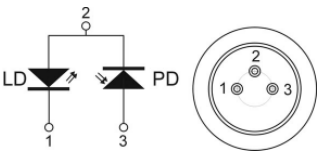
02 Customer`s Specifications

Wavelength:	780.24 nm	**Output Power:**	80.0 mW
Coarse Tuning:	765-797	**Mode Hop Free Tuning:**	-

03 Laser Diode

Article Number:	#LD-0780-0100-AR-1
Serial Number:	30928

Laser Diode Pinning



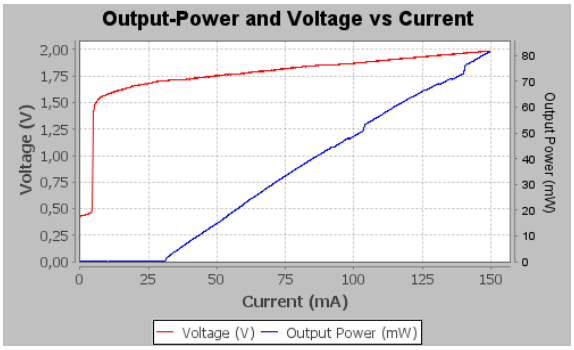
Pin1: LD Cathode
Pin2: Common, GND
Pin3: PD Anode

Negative Laser Diode
(Bottom View)

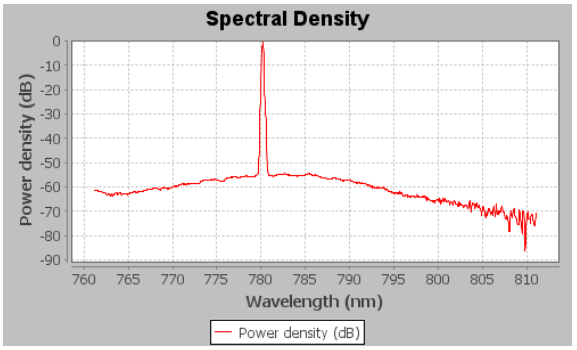
04 Stabilized Diode Laser

Wavelength:	780.2 nm	**Lasing Threshold:**	30 mA
Resonator Type:		**Slope Efficiency:**	0.678 W/A

Output Power vs. Current Characteristics



Optical Spectrum (RBW = 0.05 nm)



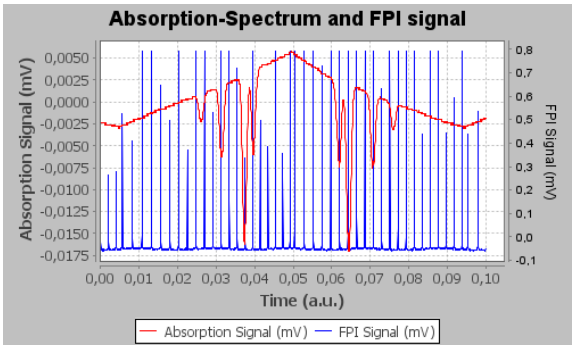
Factory Settings:

Set Power:	70.7 mW
Set Current:	136.0 mA
Set Temperature:	20.0 °C
Mode Hop Free Tuning:	21 GHz
Scan Amplitude:	46 V
Scan Offset:	73 V

Maximum Values:

Maximum Power:	81.3 mW
Maximum Current:	150 mA
Maximum Voltage:	2.28 V
Maximum MHFTR:	52 GHz @ 96.0 mA

FPI Signal (FSR= 1GHz)

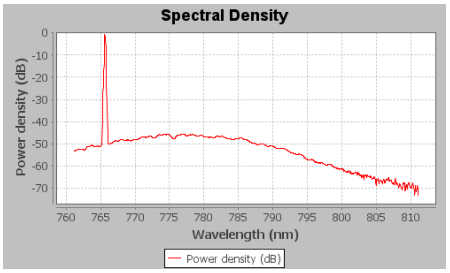


Power values are measured behind the integrated optical Isolator.

05 Coarse Tuning

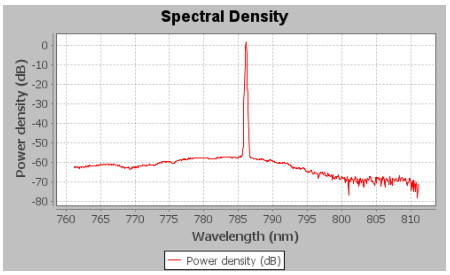
Minimum Wavelength: 765.6 nm
Lasing Threshold: 60 mA
Temperature: 20.0 °C
Maximum Power: 48.7 mW
Maximum Current: 147 mA

Optical Spectrum (RBW = 0.05 nm)



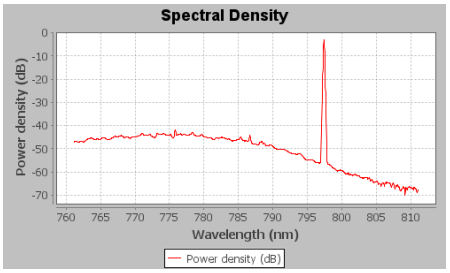
Gainmaximum: 786.1 nm
Lasing Threshold: 29 mA
Temperature: 20.0 °C
Maximum Power: 82.8 mW
Maximum Current: 147 mA

Optical Spectrum (RBW = 0.05 nm)



Maximum Wavelength: 797.4 nm
Lasing Threshold: 91 mA
Temperature: 20.0 °C
Maximum Power: 31.5 mW
Maximum Current: 147 mA

Optical Spectrum (RBW = 0.05 nm)

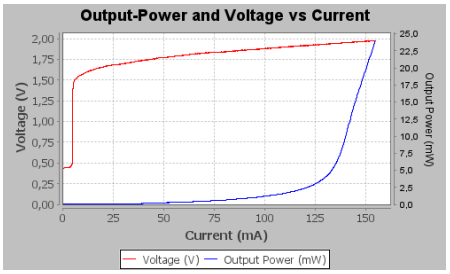


Power values are measured in front of integrated optical Isolator.

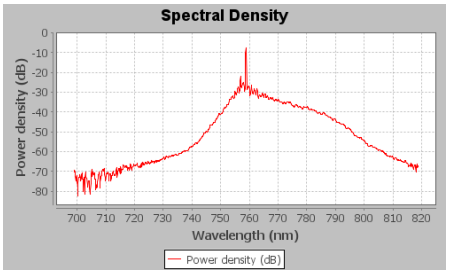
06 Free Running Laser Diode

Wavelength: 758.8 nm
Lasing Threshold: 124 mA
Maximum Power: 23.9 mW
Maximum Current: 155 mA
Temperature: 20.0 °C
Slope Efficiency: 0.783 W/A

Output Power vs. Current Characteristics



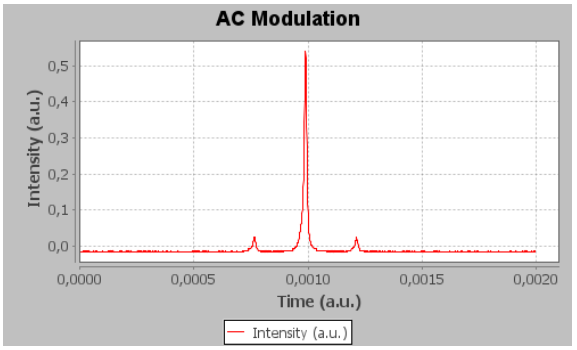
Optical Spectrum (RBW = 0.05 nm)



Power values are measured in front of the integrated optical Isolator.

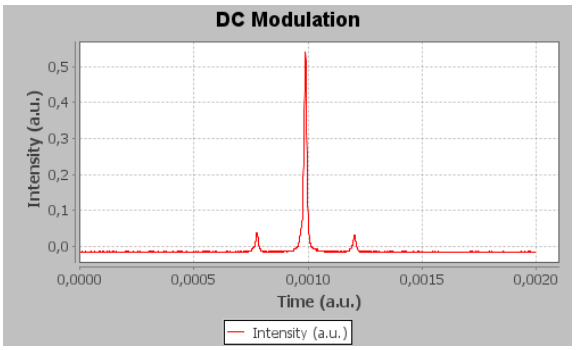
07 RF-Modulation (AC coupled)

Modulation Frequency: 20 MHz
RF Power (50 Ω -Source): 5.0 dBm



08 RF-Modulation (DC coupled)

Modulation Frequency: 20 MHz
RF Power (50 Ω -Source): 4.0 dBm



09 Integrated Isolator

Isolator: #OK-000548 **Serial Number:** 849

10 Integrated Fiber Coupling

Fiber Coupling not installed

11 Laser Optics

Collimator Lens:	#OK-000520	**Optical Grating:**	#OK-000347
Beam Steering Mirror:	yes		
Integrated Beam Shaping:	None		

12 Electronics / Software

DC 110/19" :	2176
DC 110 Mon :	2499
DCC 110/0,5A :	2236
DTC 110 30W :	3005
SC 110 :	2346
DCB 110 :	1608

13 Jumper Settings Plug-in Modules

DCC 110:	External input via DA# 0 backplane line
SC 110:	Feed forward via DA# 0 backplane line
SC 110:	External input not set

14 REACH Information

Be aware this laser system contains candidates for "Substances of very high concern" (SVHC) and will potentially be included under the authorization list under European regulation EC No, 1907/2006.
If the product is handled properly, no health risks arise during use due to the harmful SVHC contained in it.
After use, the product must be disposed of in accordance with the applicable legal regulations.
Please contact TOPTICA Photonics AG for further details.

15 Quality Control

Production:	Waginger Michael	Date: 27.03.2009
Final Check:	_____	Date: 27.03.2009